

SOFT INFRASTRUCTURE

MEDIA, FOG, AND MICRO-URBAN INTERVENTIONS

2025-2026 QIV WANG
INSTRUCTOR: RENE C DAVIDS

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Abstract

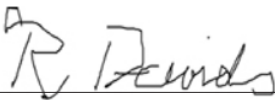
This thesis explores fog as a soft media infrastructure that extends mediated architecture beyond solid screens and fixed façades. As commercial media increasingly occupies architectural surfaces, buildings have become sites of visual consumption, often prioritizing visibility over public interaction. In response, this project asks how media architecture can become spatial, accessible, and civic rather than merely visual.

Through material experiments with fog, projection, and column-based infrastructure, the thesis develops an evanescent architecture that can receive images, occupy volume, and remain physically permeable. Unlike conventional media surfaces, gas can be entered, disturbed, and temporarily reshaped by bodies. Across a series of prototypes—from Piccadilly Circus and Manhattan to Kobe, Times Square, Taipei, Phoenix, and Hunters Point—the project tests how soft media can operate as façade, public interface, climatic device, and political platform.

The final proposal applies this system to many scenarios in urban, where permanent construction remains environmentally and politically contested. Using structural columns, fog columns, media columns, platforms, and projections, the project activates the site without fully solidifying it. It creates a temporary public space for gathering, art, environmental storytelling, and community expression, proposing media not as commercial advertisement, but as an atmospheric infrastructure for civic life.

This thesis book was produced in the context of Professor Rene C Davids' seminar and studio 204 2025-2026, in partial fulfillment of requirements for the Master of Architecture degree at the University of California, Berkeley.

Thesis Advisor Signature Here
Rene C Davids



DATE

Chair's Signature Here
Lisa Iwamoto, Chair

DATE

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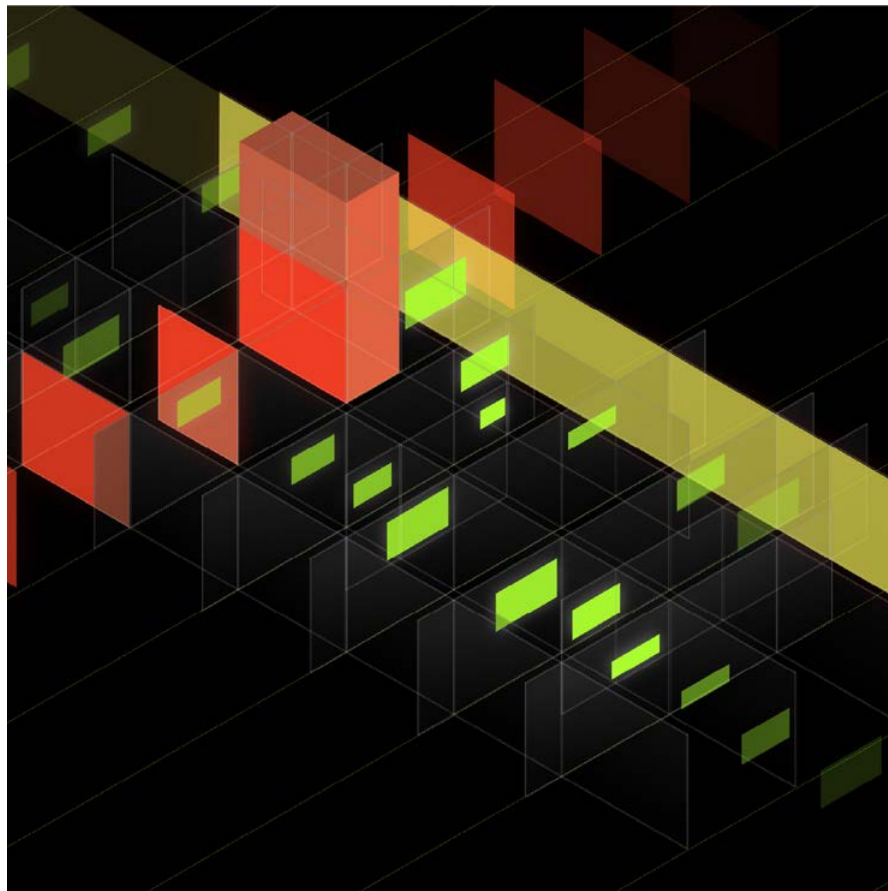
Media has long shaped the way architecture is seen, occupied, and valued. Since the expansion of commercial advertising in the 1970s, advertisements have increasingly climbed onto building surfaces, transforming walls, façades, and urban elevations into sites of visual consumption. Over time, the medium of advertisement has evolved from printed posters and static billboards to illuminated signs, electronic screens, and large-scale LED façades. Each technological shift has made media more efficient at capturing public attention, while also making architecture increasingly dependent on visibility, image, and spectacle.



This transformation has not remained limited to the exterior of buildings. As commercial and cultural spaces pursue more immersive forms of experience, media has also moved indoors. Interior walls, ceilings, and spatial surfaces are now frequently activated by posters, screens, projections, and environmental graphics. In these spaces, media no longer functions only as decoration or signage; it begins to shape atmosphere, circulation, orientation, and the way bodies move through space.



These changes have gradually influenced architectural design itself. In the late twentieth century, architects began to learn from commercial signage and the visual logic of Las Vegas, recognizing that buildings could communicate through image, symbol, and surface. Today, this relationship has become more direct and more spatially consequential. In places such as Times Square, media often occupies architecture through a crude add-on logic, covering existing façades with screens and billboards. At the same time, new buildings increasingly reserve space for media in advance, embedding screens, projection surfaces, and branded interfaces as part of the architectural design. Media is no longer simply attached to architecture after the fact; it is becoming one of the conditions through which architecture is conceived.



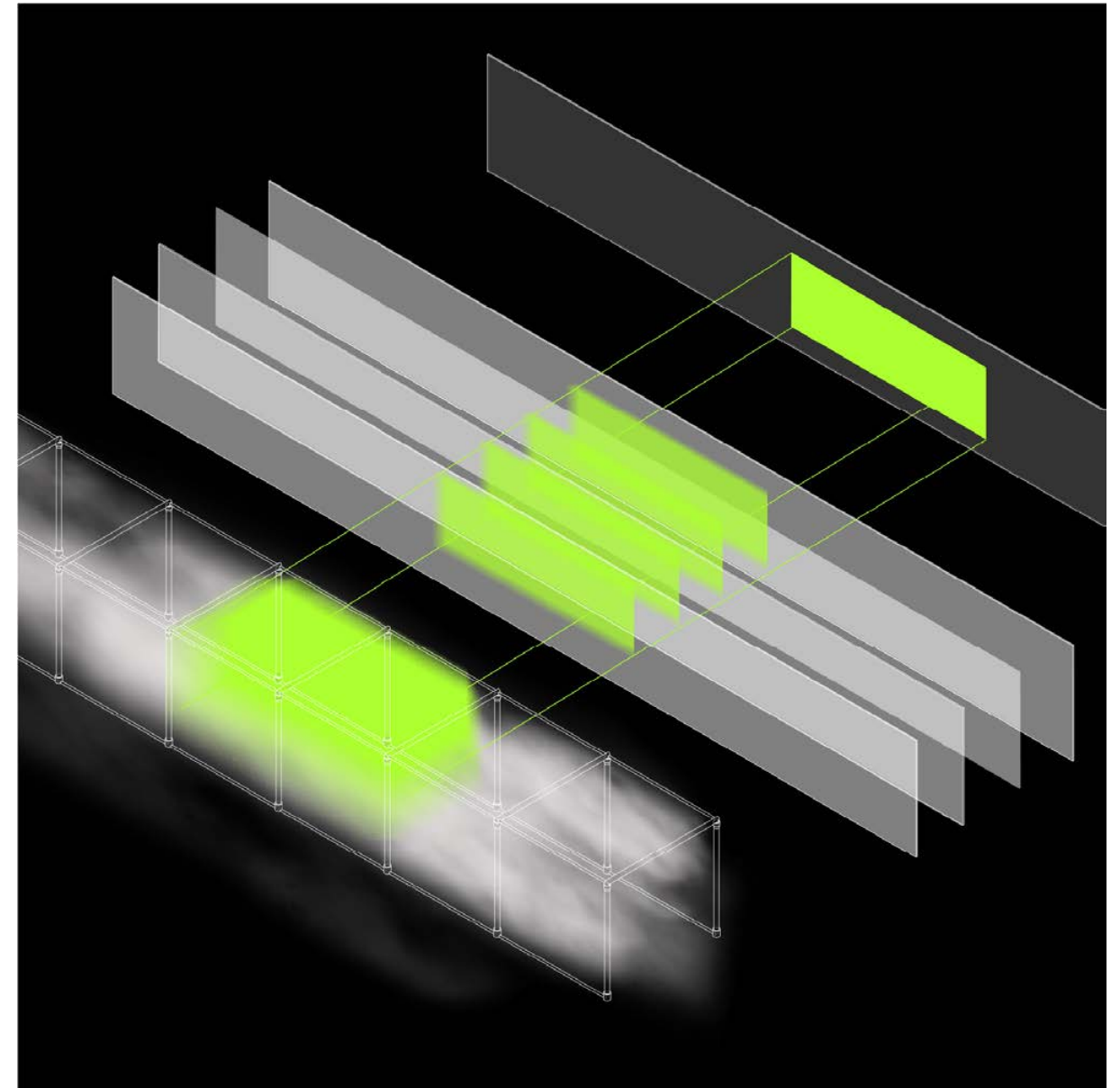
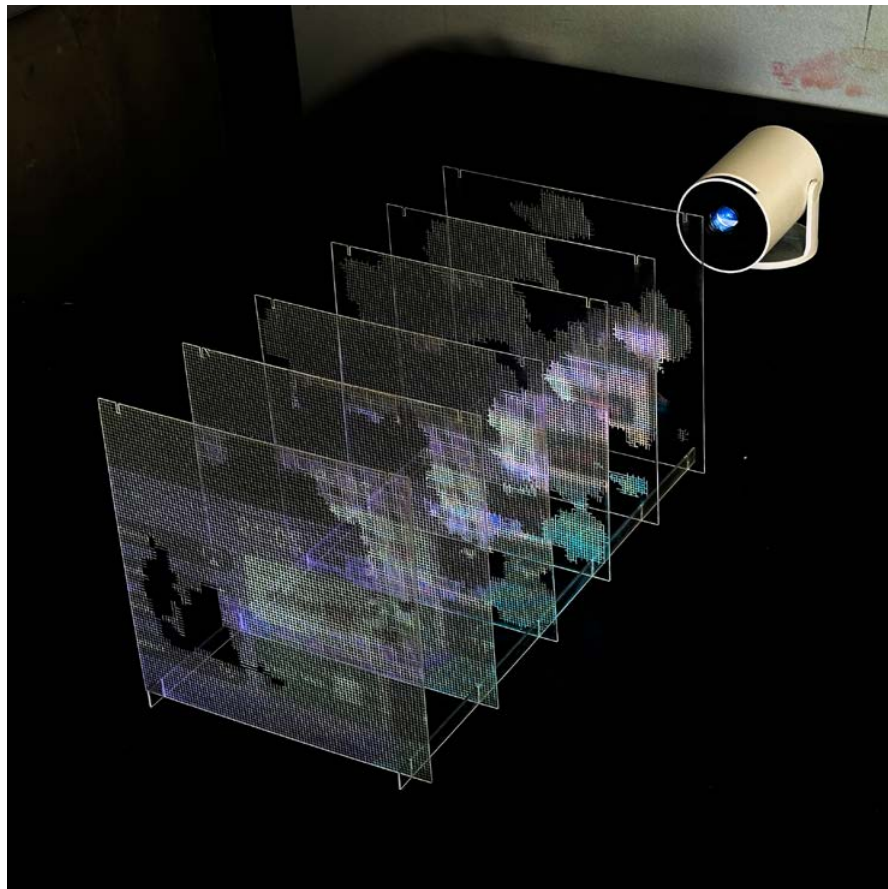
However, even as media becomes more integrated into architecture, its material logic often remains conventional. Most media architecture still relies on solid surfaces: walls, screens, panels, and façades. These surfaces are visually active but spatially closed. They display images and create immersive atmospheres, yet they rarely allow the public to pass through, touch, interrupt, or physically participate in them. Media architecture has become visually immersive, but often remains physically inaccessible.

This raises the central question:

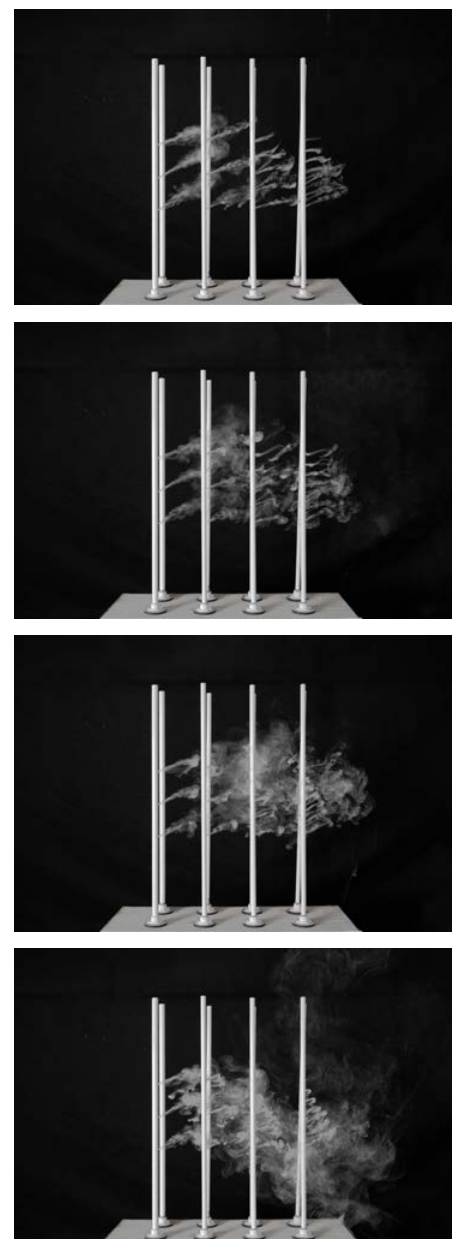
if the commercialization of urban space cannot be fully resisted, and if the media-ization of architecture cannot simply be reversed, how can architects design a form of media architecture that participates more actively in civic life?

Rather than rejecting media, this thesis asks how media might become spatial, accessible, and participatory.

This question led to the exploration of a volumetric medium. In the early phase, semi-transparent materials were tested as projection surfaces, arranged into a matrix, and activated with projected images. Although this created a more layered and porous visual field, the medium remained materially solid. People could look through it or move around it, but they still could not move through the medium itself.

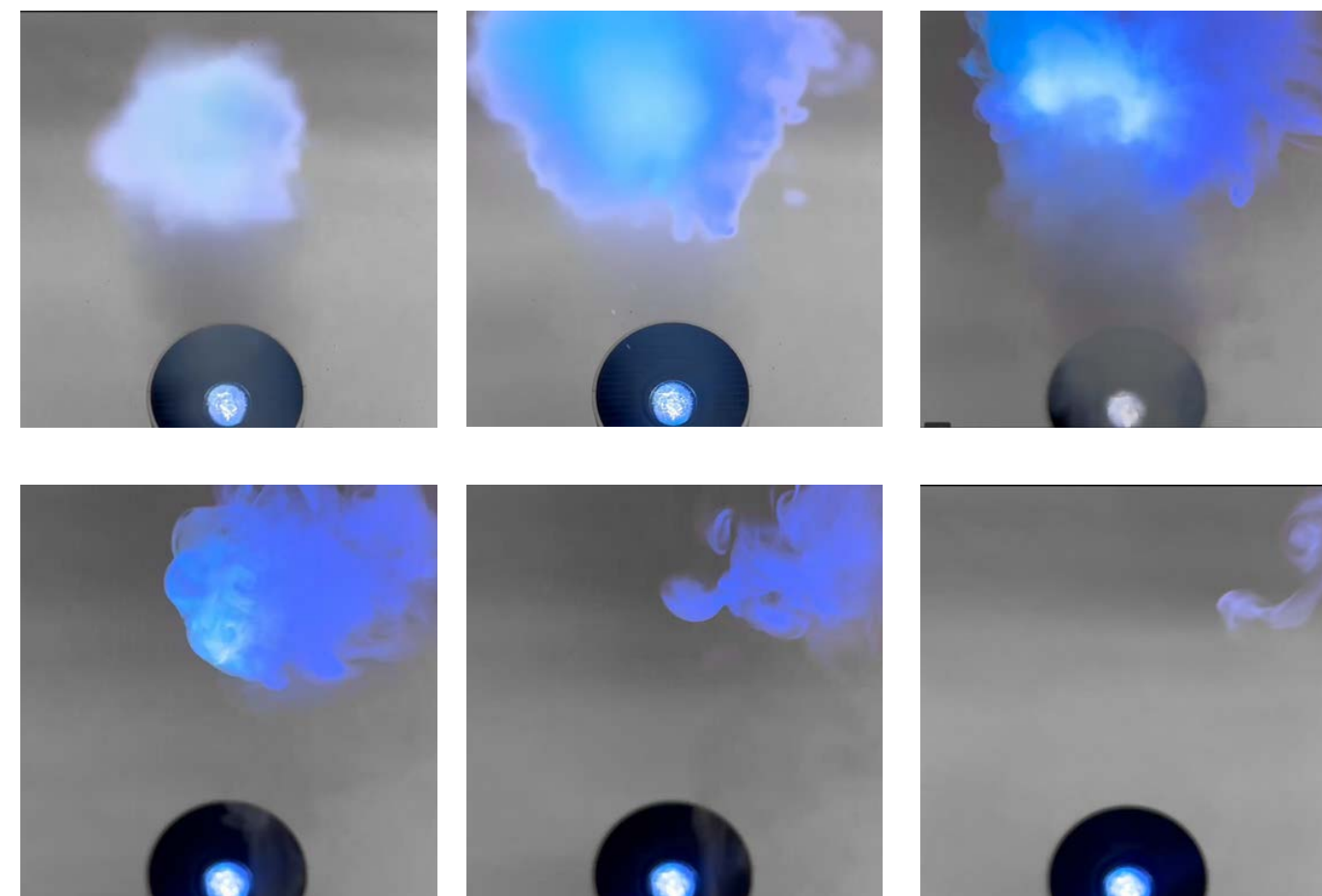


This limitation led me to **gas**. By projecting onto particles suspended in the air, media could move beyond the flat screen and become a three-dimensional volume: visible, atmospheric, and physically accessible.



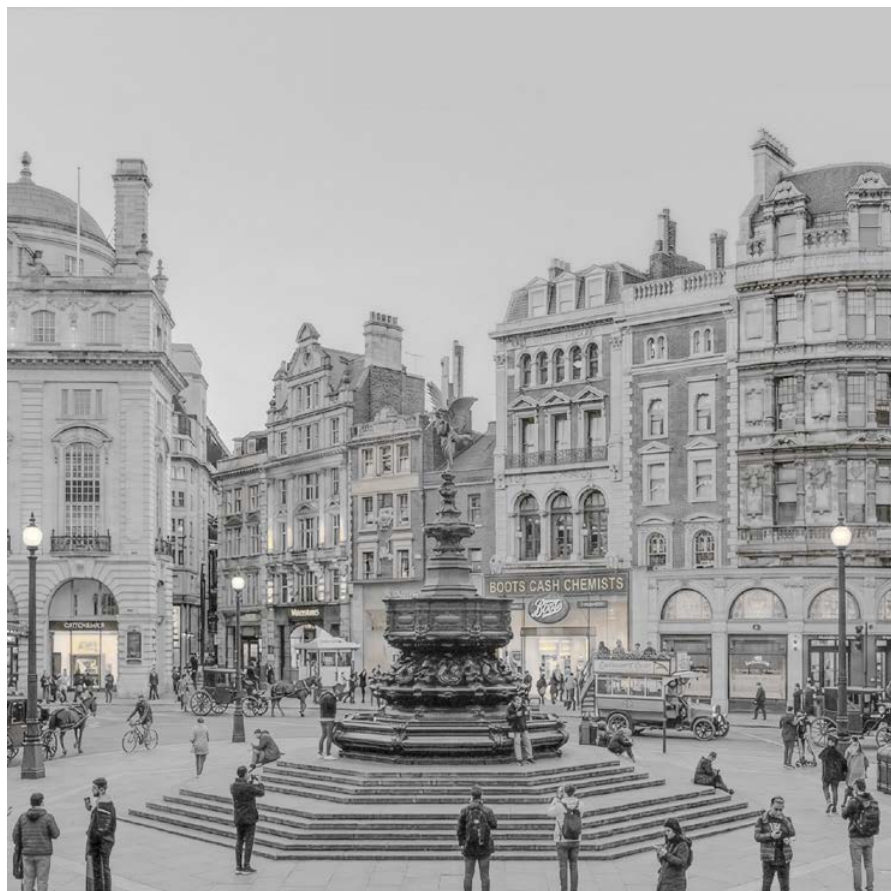
Experiment 01: Fog

The next step was to test how the shape of this atmospheric medium could be controlled. The study began by examining installation artists who use arrays of columns, vents, or nozzles to regulate the release of fog and define its overall volume. Following this logic, a prototype using a field of fog columns was developed to test whether the medium could be shaped through distribution and pressure. The experiment suggested that, with precise control of nozzle placement and air pressure, fog can be organized into a legible volumetric form rather than remaining a formless atmosphere.



Experiment 02: Projection

A second experiment tested fog as a projection medium. Using a projector and a fog machine, I projected light into a cloud of suspended water particles. Through the Tyndall effect, the projected image became visible within the fog, producing a media volume rather than a flat image surface. This test confirmed the potential of fog to operate as a spatial medium: it can receive projection, occupy volume, and remain physically permeable.



After testing the feasibility of fog as a projection medium, the potential applications of this soft media infrastructure were explored. The first possibility was to use fog as an alternative to existing electronic screens. For this test, Piccadilly Circus in London was selected, where historically significant building façades have been largely covered by commercial media surfaces.



The proposal replaces the fixed electronic screen with a temporary fog medium. Instead of permanently attaching media to the façade, fog allows the media surface to appear only when needed. When projection or advertising is active, the fog becomes a visible volumetric screen. When media is no longer required, the fog can dissipate, allowing the original architectural surface to reappear. In this way, the prototype protects the façade while still accommodating the demand for urban media.



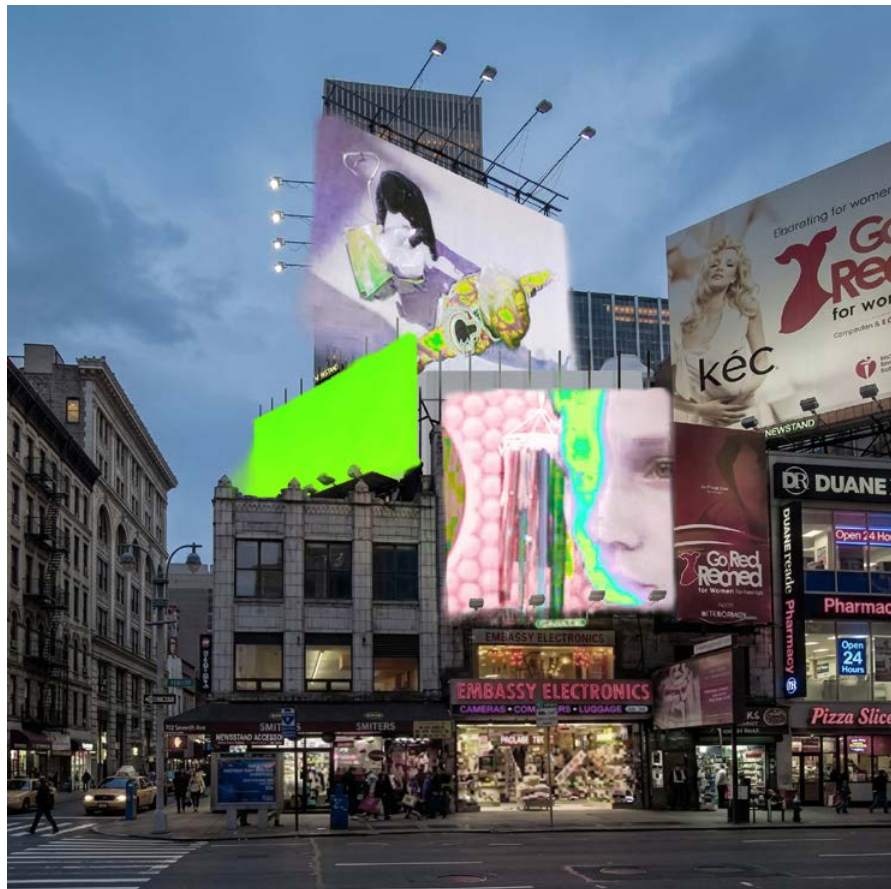
model: The crop of Piccadilly Circus facade with soft infrastructure



model: The crop of Piccadilly Circus facade with soft infrastructure



The same soft infrastructure was then tested in Manhattan, replacing selected billboards with fog-based media surfaces. This prototype suggested another limitation. Although fog offered a softer and more temporary alternative to the electronic billboard, the intervention still operated mainly as a replacement for a flat media surface. It changed the material of the screen, but not the spatial relationship between media, architecture, and the public.



This revealed that the value of fog should not be limited to making a softer billboard. If fog is a volumetric medium, then the project should use its depth, permeability, and spatial instability rather than forcing it back into the logic of a two-dimensional screen.



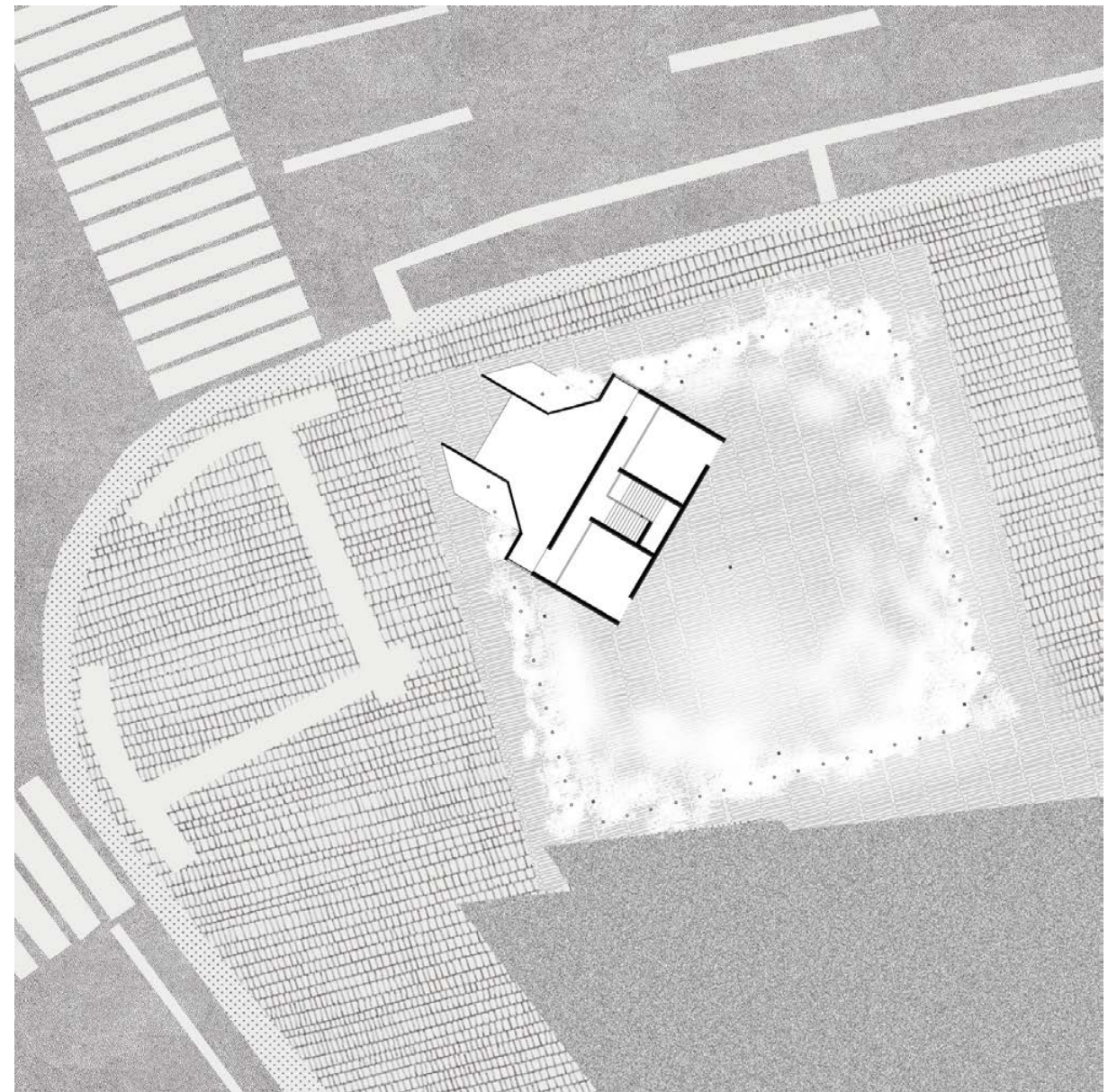
model: The crop of Manhattan shops facade with soft infrastructure



This led to a new question: what would happen if this strategy were integrated into the design of a new building from the beginning, rather than applied afterward as a replacement for existing media surfaces?



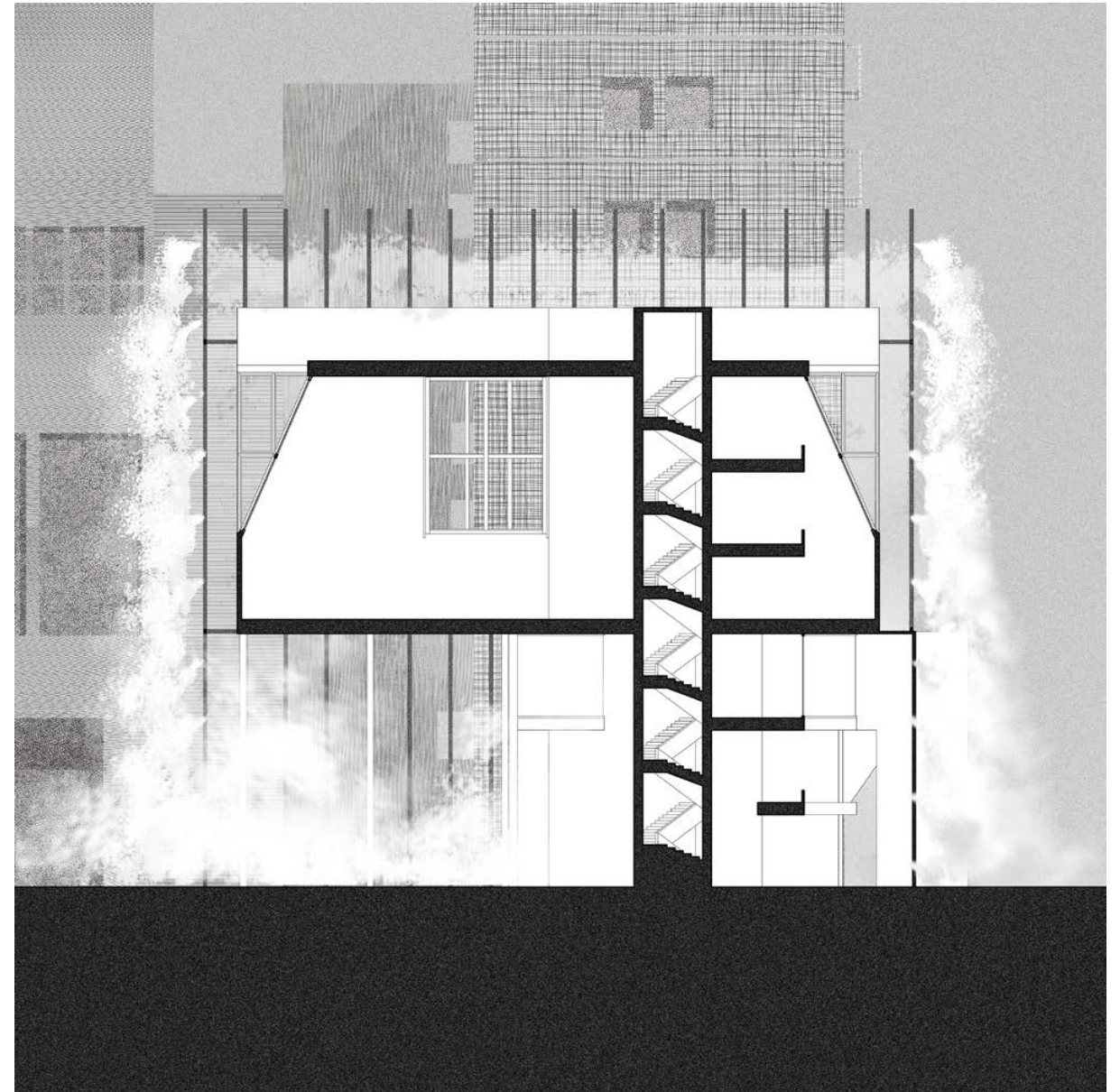
The next prototype is located in an art district in Kobe, Japan. The project integrates soft media infrastructure into a hybrid program of art offices and exhibition spaces. Instead of attaching media to a finished façade, fog becomes the building's outer atmospheric layer. It wraps around smaller solid architectural cores, producing the appearance of a soft, cloud-like cube. Branding, cultural content, and projected images can then be displayed through this temporary and permeable medium.



In plan, the boundary between interior and exterior becomes intentionally blurred. The fog layer opens the ground floor and creates a more accessible public interface between the building and the street.



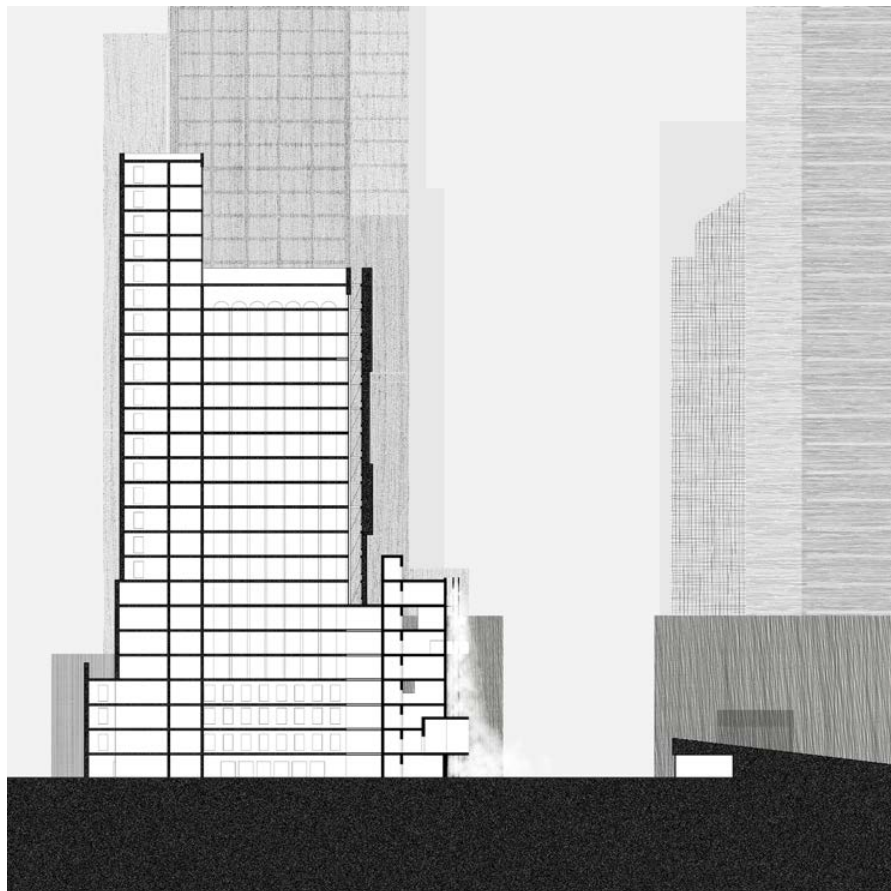
Model: The crop of a corner of the building



In section, the outer envelope becomes difficult to define as a conventional architectural surface. It has visual presence, but no fixed material boundary. Rather than a façade, it operates as a soft media volume that can be entered, seen through, and temporarily activated.



The research then extended the same strategy to Times Square, a site already saturated with solid media surfaces. Rather than replacing one billboard at a time, this prototype asks what might happen if the district's hard media environment were reimagined through soft infrastructure.



In this test, the mediated façade begins to lose its fixed edge. Composed of fog, projection, and temporary atmospheric layers, the façade becomes blurred and volumetric. It is no longer a flat image surface attached to architecture, but a thickened media field that expands outward into urban space.



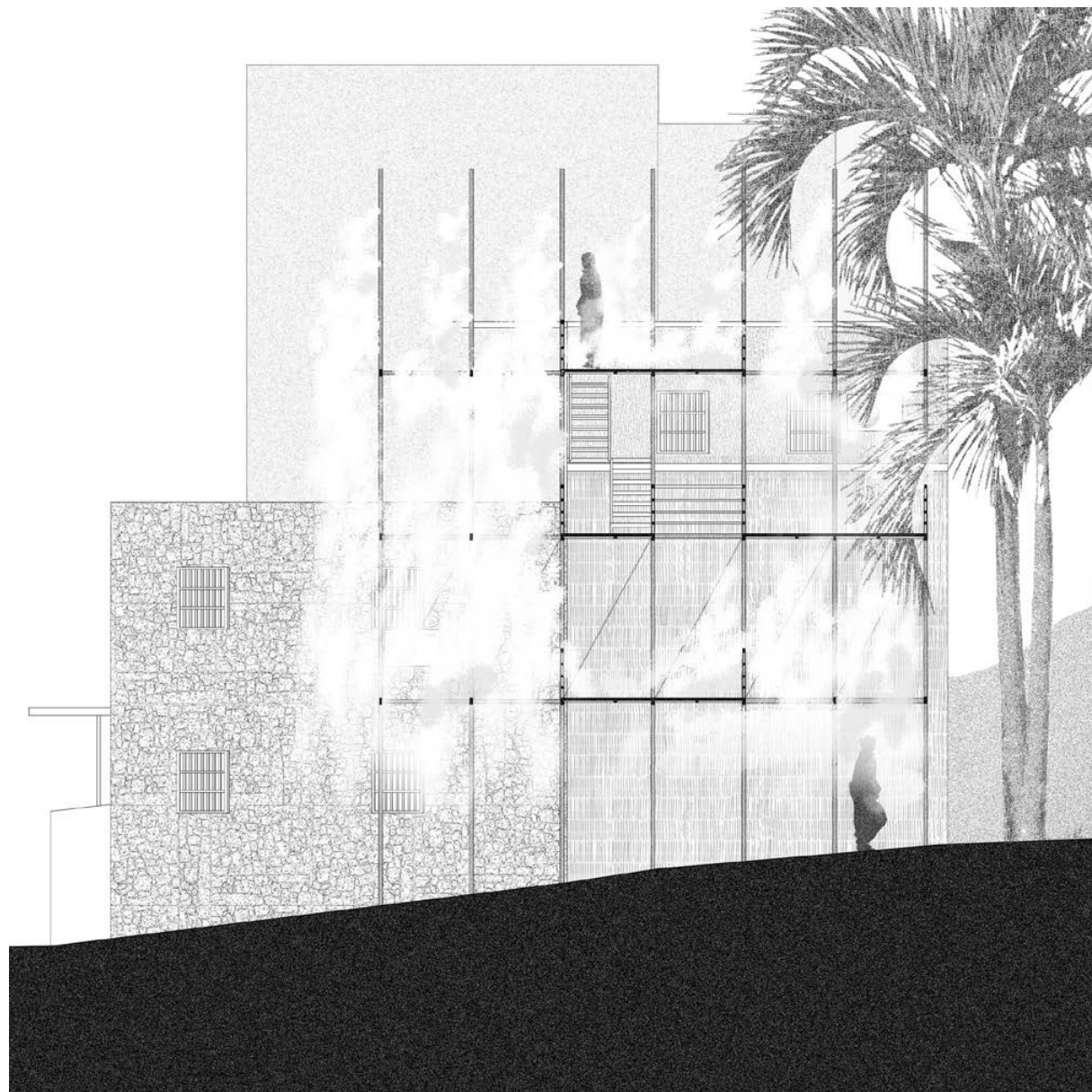
Model: The crop of the building



At a larger urban scale, I began to ask whether this medium could operate not only as a media surface, but also as an environmental public infrastructure. This prototype is located in Taipei, Taiwan, where dense residential neighborhoods often experience intense heat and limited access to nearby public facilities. In many aging communities, elderly residents leave their homes during the day to search for cooler public spaces. However, these facilities are not always close to where they live. At the same time, informal residential extensions have created narrow gaps between buildings—spaces that are usually overlooked, but could support small-scale interventions.



The proposal uses these residual gaps to install soft infrastructure. Fog can provide localized cooling, while projection turns the infrastructure into a recognizable public signal. Rather than functioning as commercial advertisement, media is used here as a civic branding strategy: a way to make small cooling spaces visible, legible, and collectively recognized across the city. Through a network of micro-interventions, the project imagines fog as both a thermal device and a public media system.



Section: Micro-urban intervention



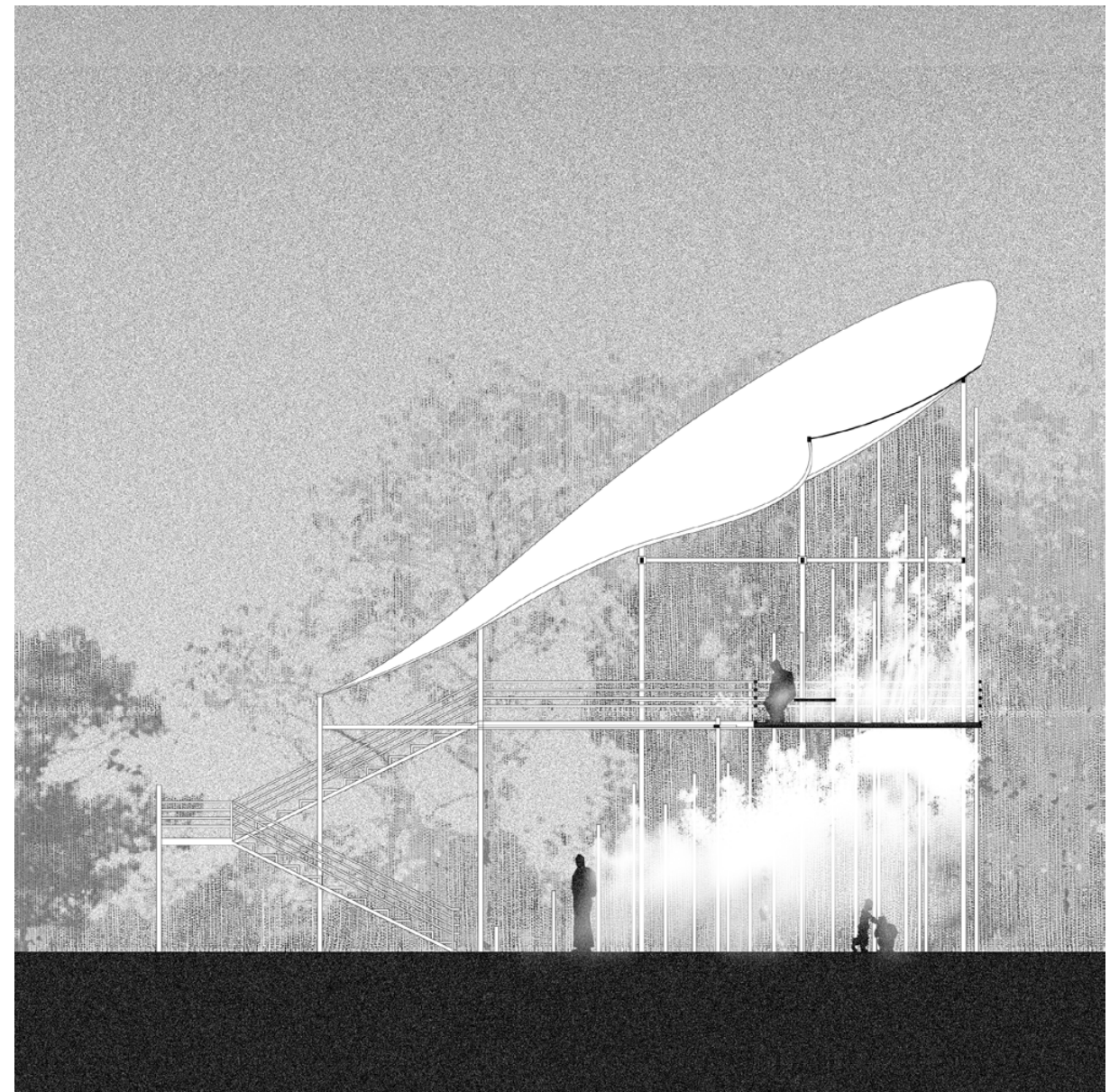
Model: Micro-urban intervention



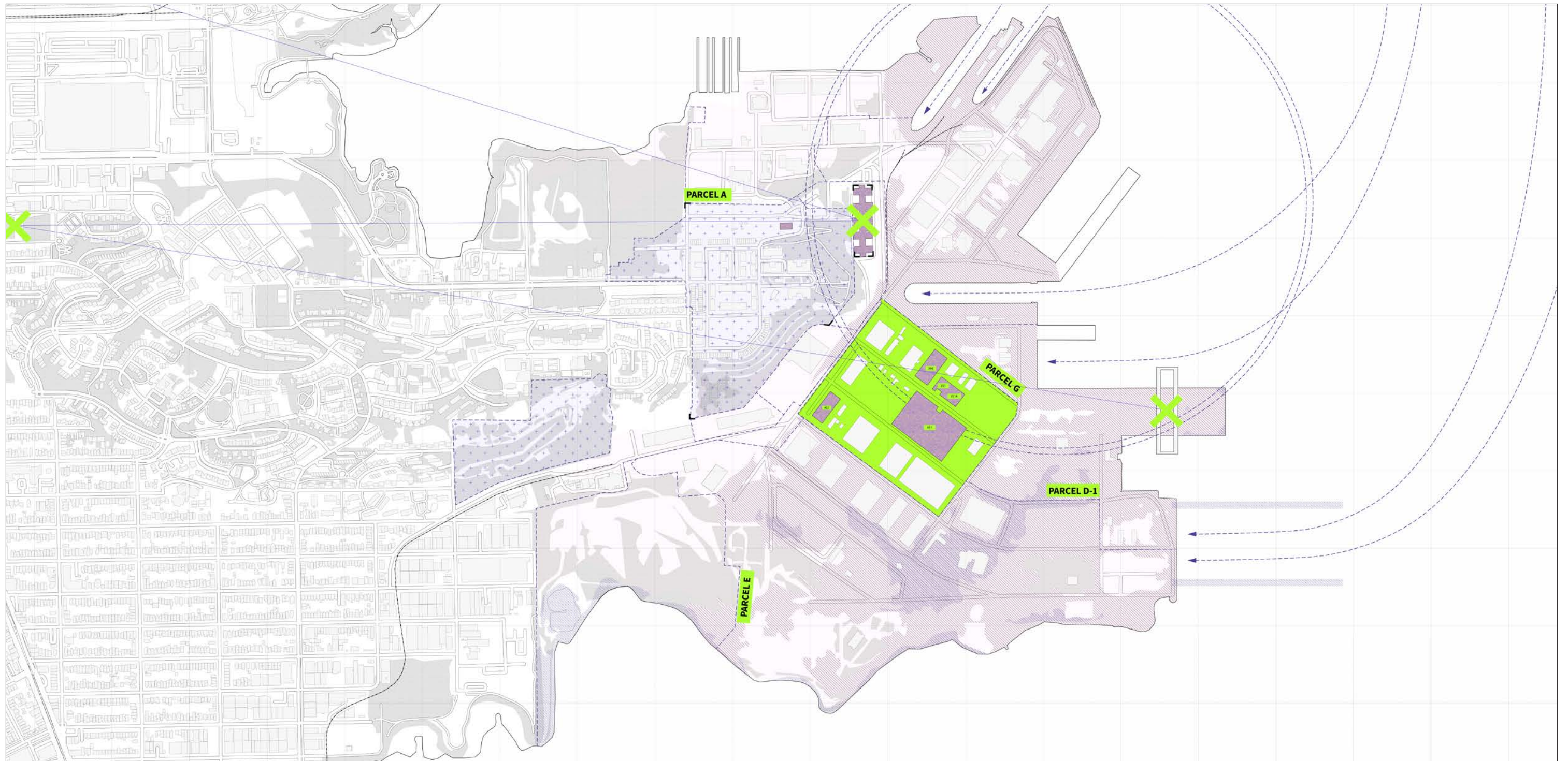
A second environmental prototype tests this infrastructure in Phoenix, Arizona. In the afternoon, when solar exposure and outdoor temperatures are most intense, public parks often become difficult to use. By combining fog-based soft infrastructure with public amenities, the project attempts to extend the period of outdoor activity and make open space more inhabitable during extreme heat.



In this prototype, media is used in a more interactive way. Rather than projecting fixed images or branding content, the system visualizes thermal conditions in real time. Changes in temperature, shade, humidity, or occupancy can affect the density, color, or intensity of the projection, allowing visitors to perceive the microclimate around them. Here, media becomes a tool for environmental awareness as well as public comfort.



Section: A fog-based soft infrastructure with public amenities



Finally, the research turned toward a more political application of this soft infrastructure. The site is Hunters Point in San Francisco, where a long history of environmental contamination, redevelopment pressure, and conflict between the community, government agencies, and private developers has made permanent construction highly contested. In this context, solid architecture is not only a

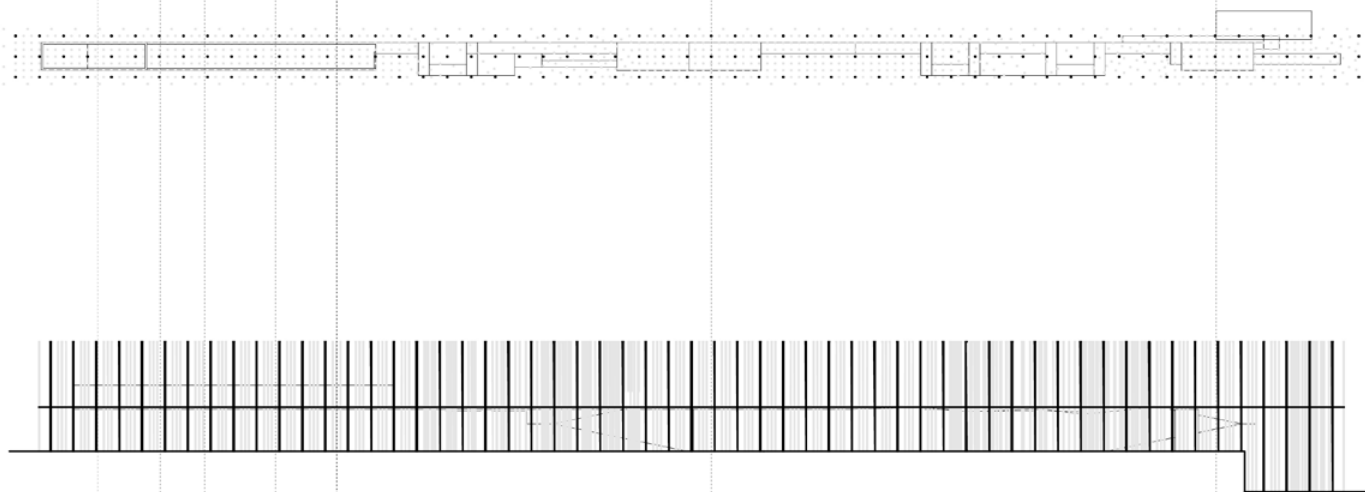
material intervention, but also a political claim—one that can easily intensify existing tensions around land, responsibility, and trust. This prototype asks whether an evanescent architecture could intervene in such a suspended condition. When the fog and projection are not active, the project exists only as a field of lightweight columns and nozzles.

It does not operate as a conventional building mass or permanent façade. When activated, however, fog produces a temporary spatial presence that can host public events, media projection, gathering, and environmental storytelling. The goal is not to resolve the conflict through architecture alone, but to create a provisional public platform that can make the site visible, usable, and

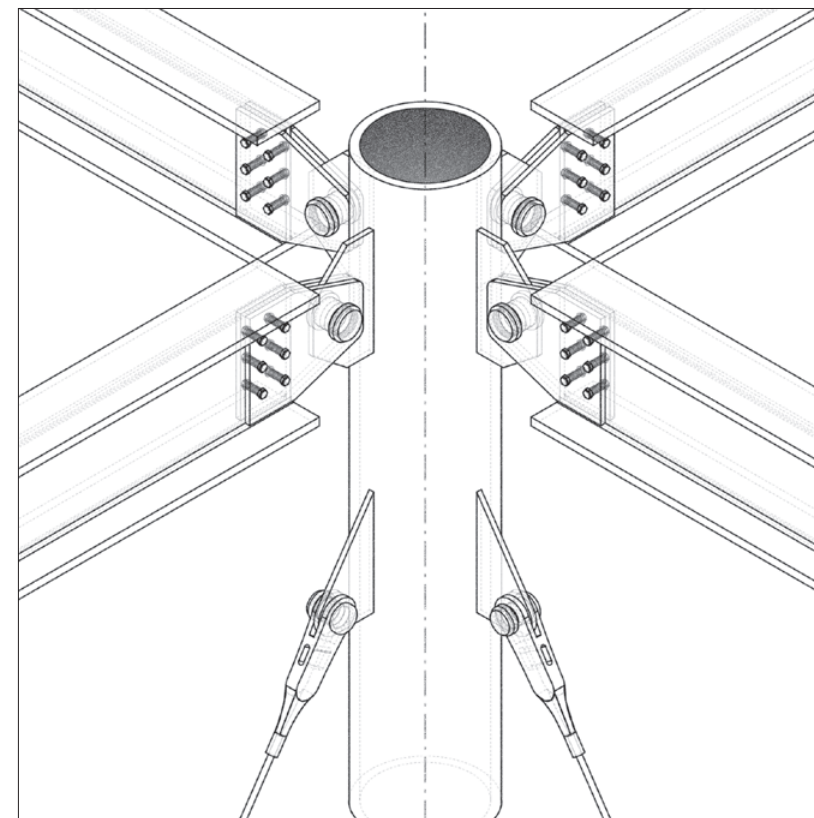
discussable before permanent development becomes possible. By activating the site without fully solidifying it, the project attempts to soften the stalemate between community, government, and development interests.



columns generating fog



structural columns in the array



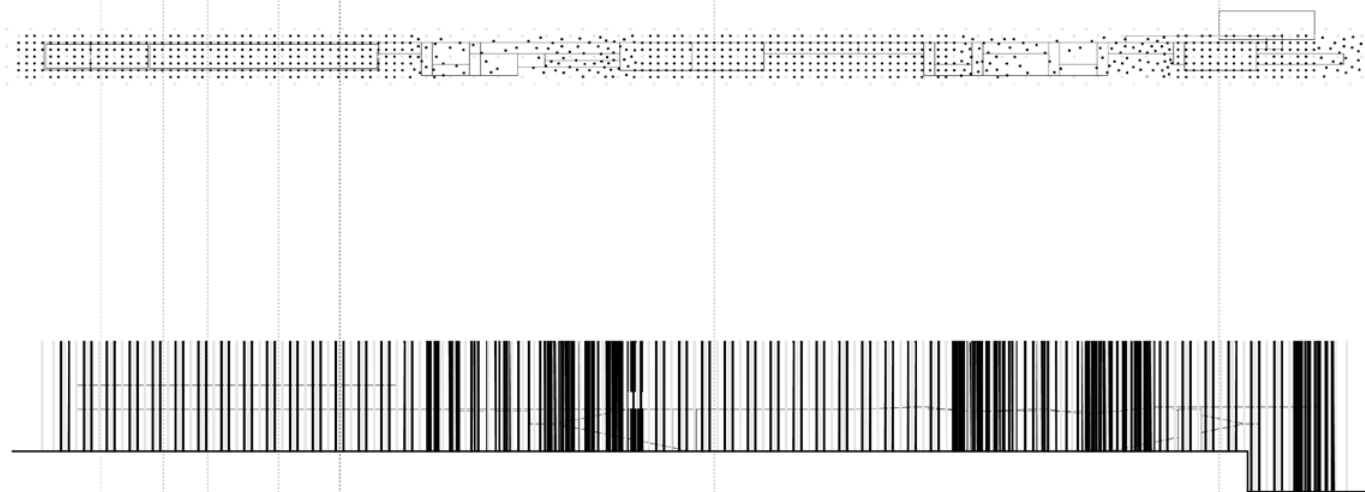
The column system is divided into three categories.

01 Structural column

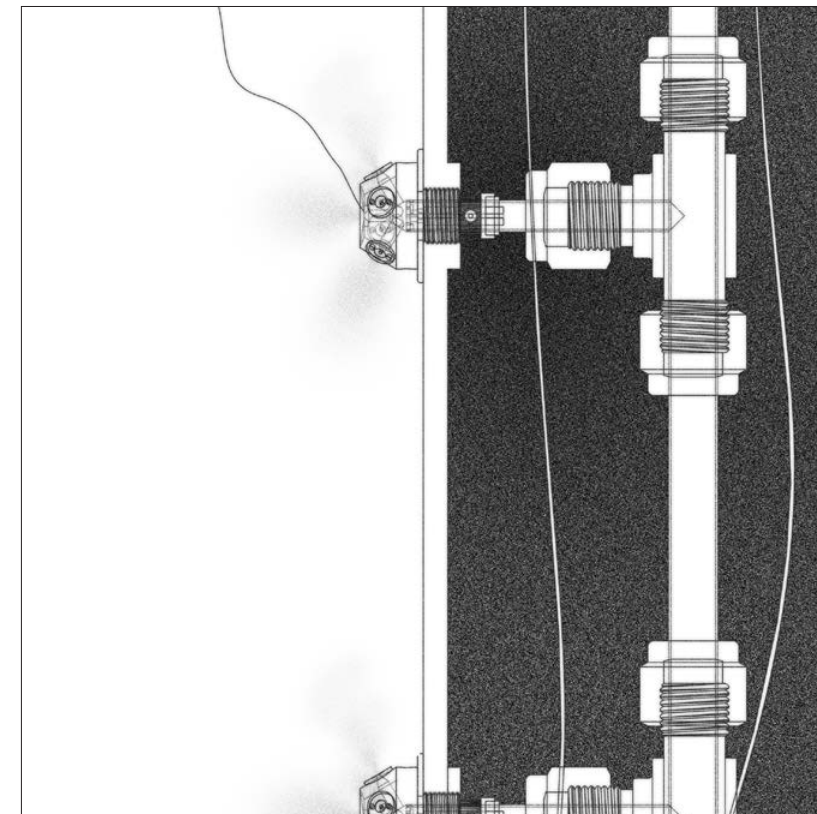
Arranged in a regular grid, these continuous vertical members support the elevated platforms and establish the project's spatial order. Each column connects to the ground through a pinned base, allowing vertical load transfer while reducing bending moment at the foundation. Lateral and shear forces are resisted by cable bracing between columns, creating a lightweight structural frame that anchors the atmospheric intervention while keeping the ground level open.



infrastructure with media projected



fog columns in the array

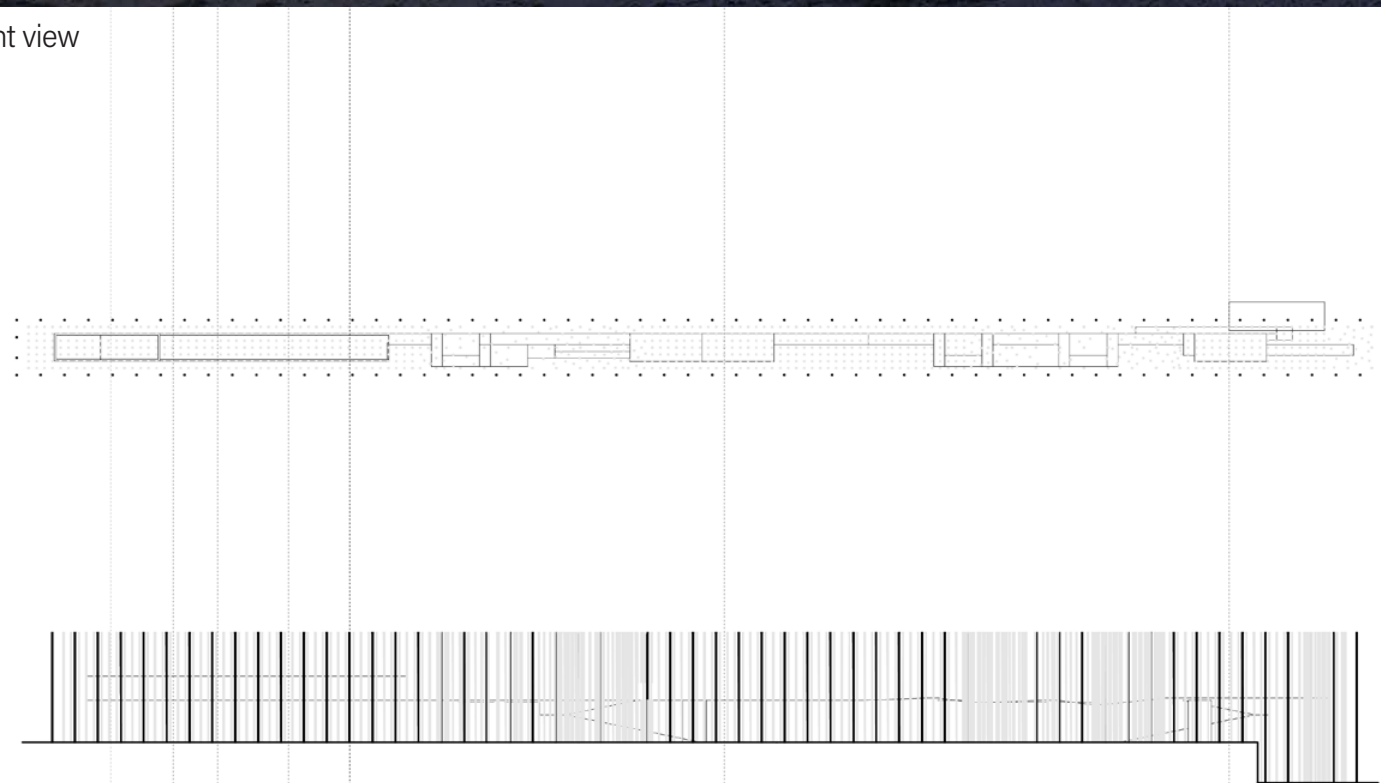


02 Fog columns

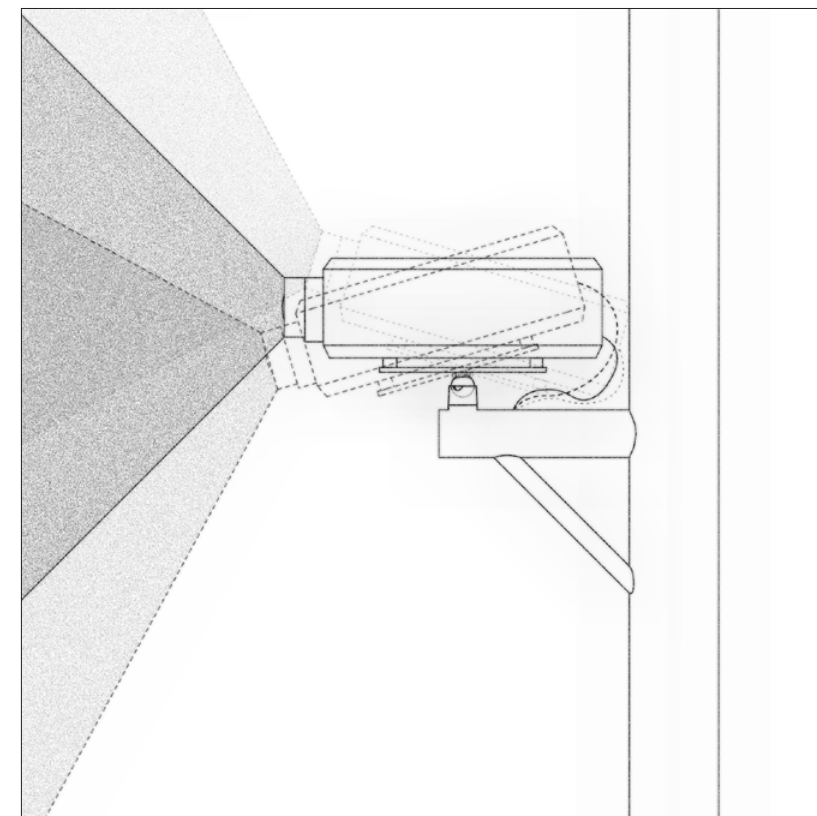
Distributed irregularly across the site, these columns generate the project's atmospheric field. Water from underground storage tanks is pumped through pipes embedded within each column and released through integrated nozzles as fine atomized mist. If needed, non-toxic fogging additives can be used to stabilize the effect. Together, the fog columns create a controllable media volume that can receive projection, cool the site, and temporarily transform the public space.



night view

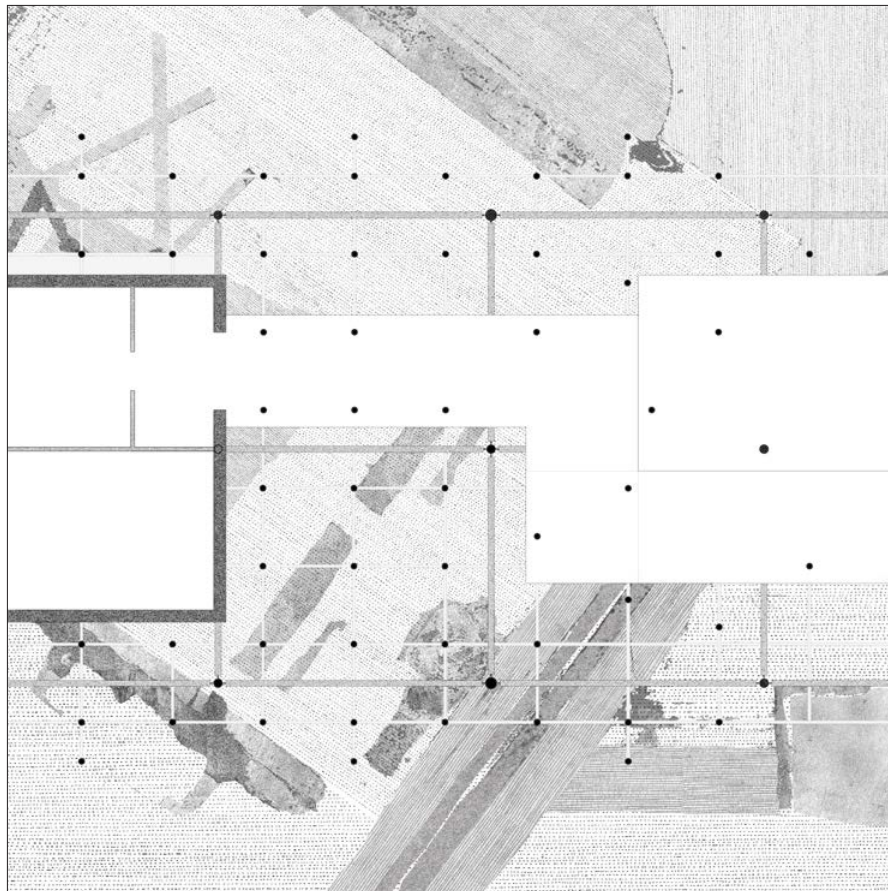


media columns in the array



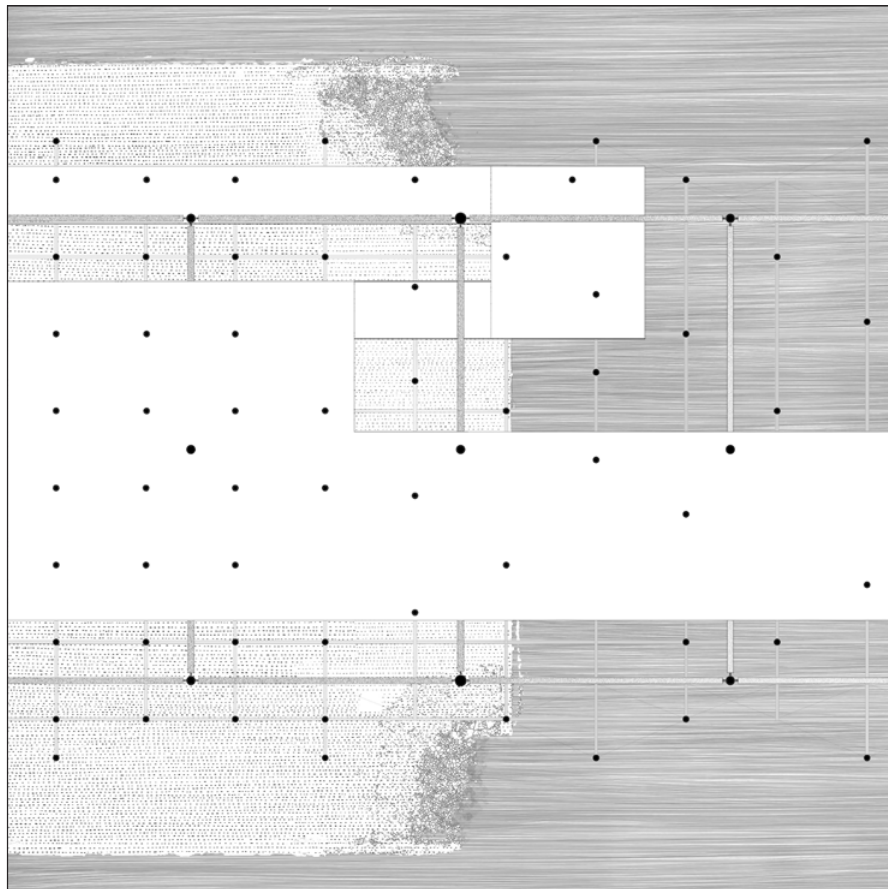
03 Media columns

Positioned along the outer edge of the field, these columns carry large-scale projectors and media equipment. From the perimeter, projection is cast inward onto the fog volume, allowing images, text, and environmental data to appear within the atmosphere rather than on a fixed surface.

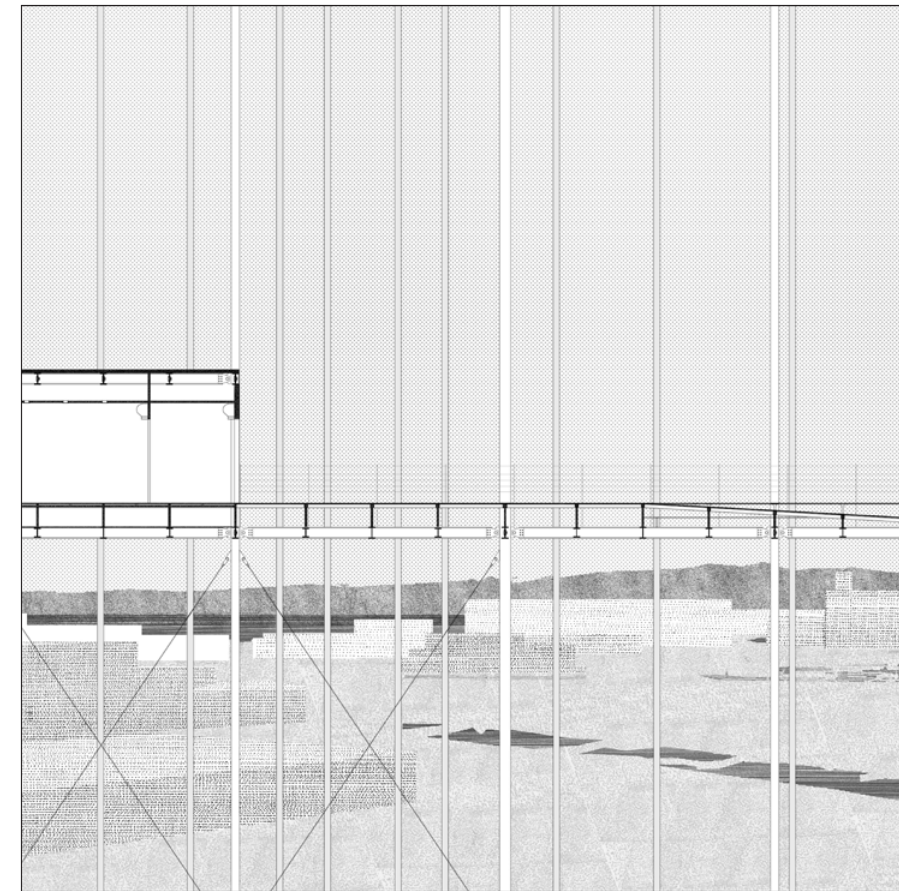


This proposal allows the site to be used without being fully built upon. Through temporary activation, the project makes the area publicly accessible while giving form to community positions, memories, and concerns. Here, media is not commercial advertisement, but a platform for public expression.

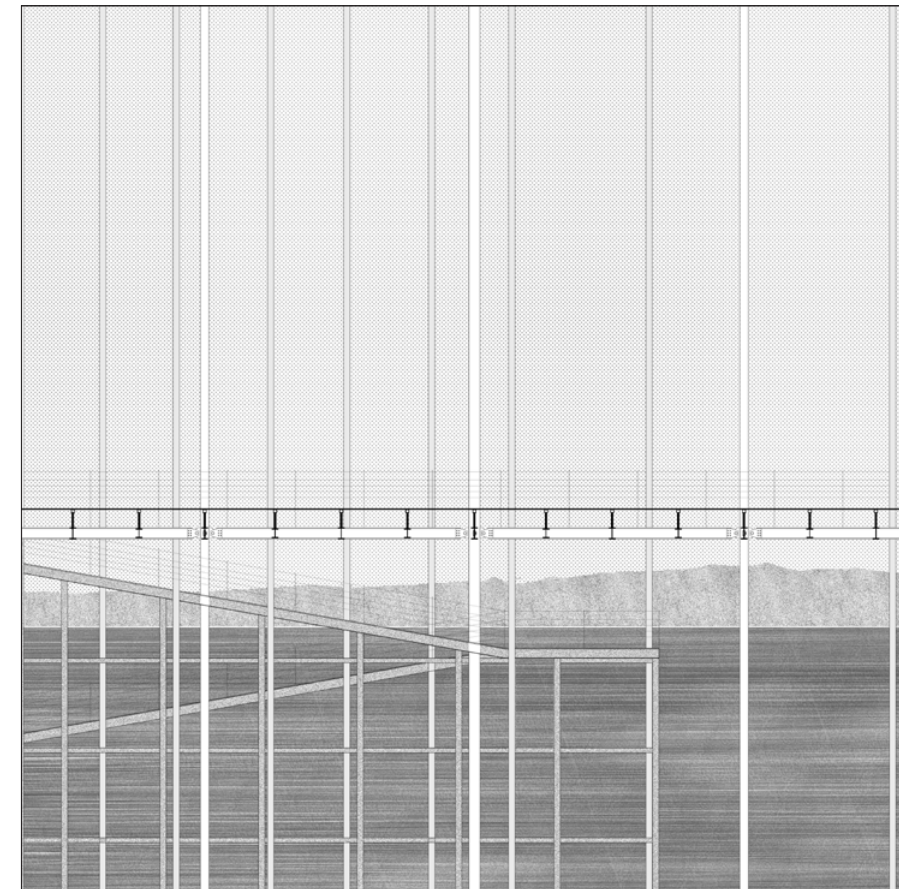
plans: the crops of the soft infrastructure on site



Because many of the currently occupied industrial buildings at Hunters Point have become artists' studios, the project treats this soft infrastructure as part of the existing art community. Stairs, platforms, fog columns, and media columns create an open framework for performance, projection, gathering, and experimental media art. Art becomes a way to challenge the site's political suspension and speak through conditions that permanent construction cannot yet resolve.



sections: the crops of the soft infrastructure on site



When fog and projection are inactive, the intervention becomes quiet and minimal. The columns remain lightly placed across the site, allowing people to focus on the existing landscape, industrial traces, and historical context. The project activates the site while respecting its unresolved condition.

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